



II Olimpiada Informática de Las Palmas

Contest – January, 26th 2024

Códigos Perdidos

Lost Codes

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Problem statement:

On the desolate planet *Zeta Prime*, intergalactic archaeologists have discovered the remains of an ancient starship destroyed by the space wars of bygone eons. Among the debris, fragments of an unknown metal inscribed with numerical sequences have been found. It is believed that these fragments are part of an ancient access code system used by the advanced civilization of *Zeta Prime*, and that each numerical sequence is actually a prime number, key to unraveling lost technologies and secrets of the cosmos.



The fragments found contain only parts of the original prime numbers.

Your mission, as a *code decipherer of the Intergalactic Federation*, is to determine from a fragment whether it is in good condition (is a prime number) or if it is necessary to reconstruct the fragment by obtaining the smallest prime number from the possible reconstructions.



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Candidate reconstructions are obtained by adding a digit (from 1 to 9, i.e., a single digit) to the right and/or left of the fragment. You must find the smallest prime number among the candidates obtained from the given fragment.

Input / Output

Input Format:

- - The input will consist of an integer number.

Output Format:

- - The output will be: ERROR, if the input is not an integer.
- - The output will be: the same number, if the input data is a prime number.
- - The output will be: the found prime number, if one exists.
- - The output will be: NON-RECONSTRUCTIBLE, if a prime number cannot be created from the code fragment.

Input /Output:

Example 1

Input:

11

Output:

11

Example 2

Input:

1003



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Output:

10037

Example 3

Input:

3431

Output:

23431

Example 4

Input:

109947

Output:

NO RECONSTRUIBLE

Example 5

Input:

**+

Output:

ERROR